Chapter 43 Review

Campbell Biology, AP Edition, 12e

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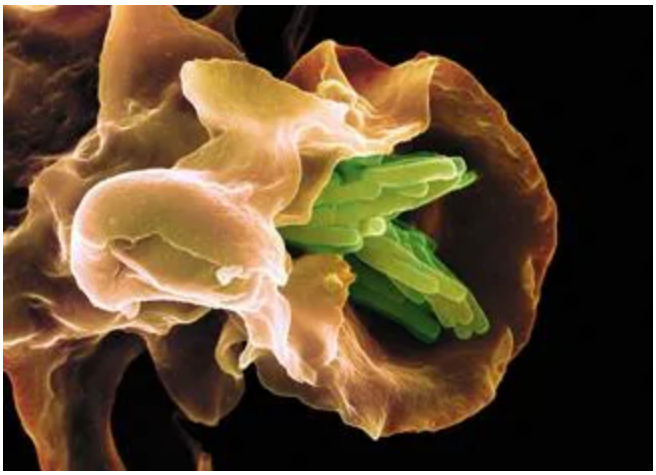


Summary of Key Concepts

Consider an amoeba feeding upon microbes as the ancient precursor of the macrophages that are now part of the innate immune response. In this context, why would “nonself” recognition be an important characteristic of those early prototypic “macrophages”? **(BIG IDEAS 1 & 2)**

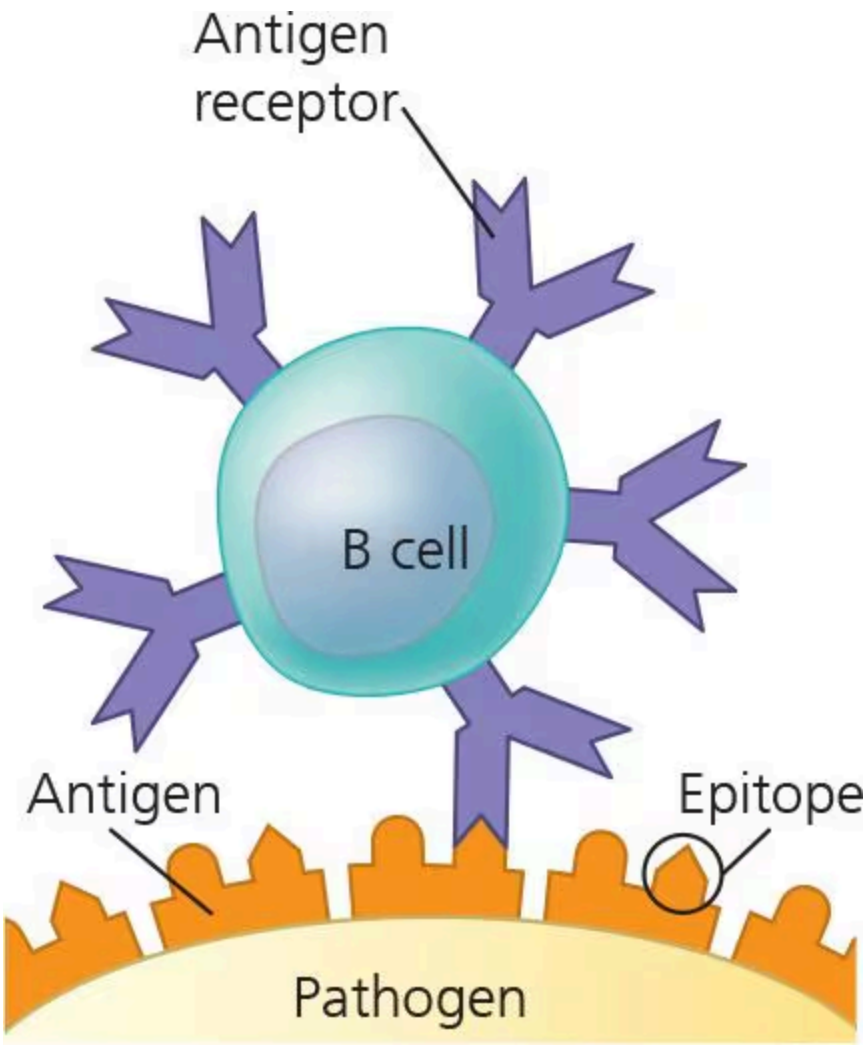
Concept 43.1: In innate immunity, recognition and response rely on traits common to groups of pathogens

- ****Innate immunity** is mediated by physical and chemical barriers as well as cell-based defenses. Recognition proteins in innate immunity are specific for broad classes of ****pathogens**. Pathogens that penetrate barrier defenses are ingested by phagocytic cells, which in vertebrates include ****macrophages** and ****dendritic cells**. Additional cellular defenses include ****natural killer cells**, which can induce the death of virus-infected cells. ****Complement system** proteins, ****interferons**, and other antimicrobial peptides also act against pathogens. In the ****inflammatory response**, ****histamine** and other chemicals that are released at the injury site promote changes in blood vessels that enhance immune cell access and action.
- Pathogens sometimes evade innate immune defenses. For example, some bacteria have an outer capsule that prevents recognition, while others are resistant to breakdown within lysosomes.

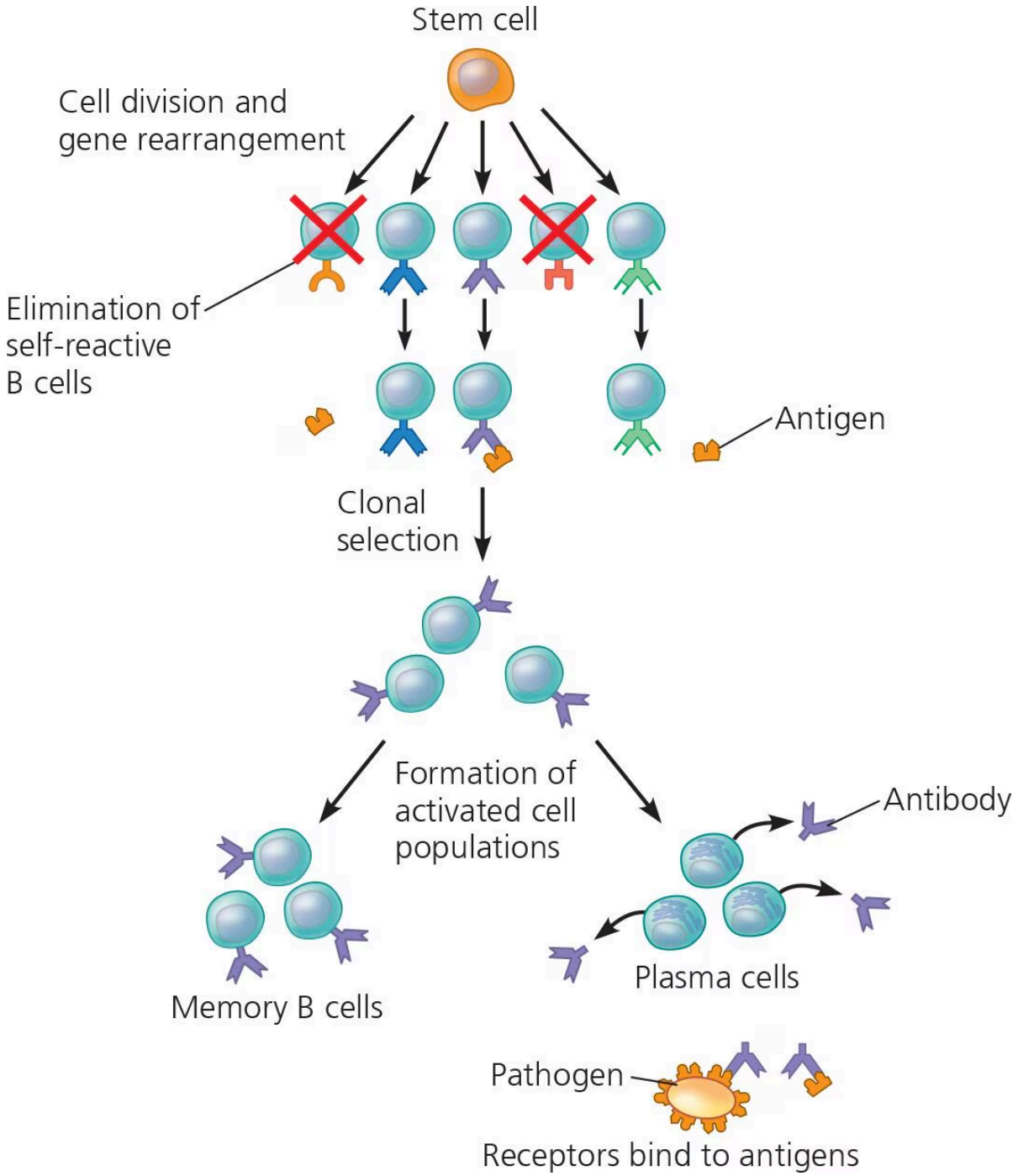
In what ways does innate immunity protect the mammalian digestive tract?

Concept 43.2: In adaptive immunity, receptors provide pathogen-specific recognition

- ****Adaptive immunity** relies on two types of ******lymphocytes:******B cells** and ****T cells**. Lymphocytes have cell-surface ****antigen receptors** for foreign molecules (****antigens**). All receptor proteins on a single B or T cell are the same, but there are millions of B and T cells in the body that differ in the foreign molecules that their receptors recognize. Upon infection, B and T cells specific for the pathogen are activated. Some T cells help other lymphocytes; others kill infected host cells. B cells called **plasma cells** produce soluble proteins called **antibodies**, which bind to foreign molecules and cells. B and T **memory cells** defend against future infections by the same pathogen.



- Recognition of foreign molecules by B cells and T cells involves the binding of variable regions of receptors to an **epitope**, a small region of an antigen. B cells and antibodies recognize epitopes on the surface of antigens circulating in the blood or lymph. T cells recognize protein epitopes in small antigen fragments (peptides) that are presented on the surface of host cells by ****major **histocompatibility complex (MHC) molecules**. This interaction activates a T cell, enabling it to participate in adaptive immunity.
- The four major characteristics of B and T cell development are the generation of cell diversity, self-tolerance, proliferation, and immunological memory. Proliferation and memory are both based on **clonal selection**, illustrated here for B cells:



Stem cell (through cell division and gene rearrangement) goes to cells with different antigen receptors on surface. Elimination of self-reactive B cells goes to fewer different cells that differ in antigen specificity. Only one of the cells diagrammed has an antigen receptor that binds an antigen present. That cell undergoes clonal selection, resulting in a formation of activated cell populations. The result is the formation of both memory B cells with receptors on the surface specific to the antigen present, and plasma cells that can secrete antibodies for the antigen present. Antibodies have specific receptors that can bind to the antigens on the surface of a pathogen.

Why is the adaptive immune response to an initial infection slower than the innate response?

Concept 43.3: Adaptive immunity defends against infection of body fluids and body cells

- ****Helper T cells**interact with antigen fragments displayed by class II MHC molecules on the surface of **antigen-presenting cells**: dendritic cells, macrophages, and B cells. Activated helper T cells secrete cytokines that stimulate other lymphocytes. In the **humoral immune response**, antibodies help eliminate antigens by facilitating phagocytosis and complement-mediated lysis. In the **cell-mediated immune response**, activated **cytotoxic T cells** trigger destruction of infected cells.
- **Active immunity** develops in response to infection or to **immunization**. The transfer of antibodies in **passive immunity** provides immediate, short-term protection.
- Tissues or cells transferred from one person to another are subject to immune rejection. In tissue grafts and organ transplants, MHC molecules stimulate rejection. Lymphocytes in bone marrow transplants may cause a graft-versus-host reaction.

Is immunological memory after a natural infection fundamentally different from immunological memory after vaccination? Explain.

Concept 43.4: Disruptions in immune system function can elicit or exacerbate disease

- In allergies, such as hay fever, the interaction of antibodies and allergens triggers immune cells to release histamine and other mediators that cause vascular changes and allergic symptoms. Loss of self-tolerance can lead to **autoimmune diseases**, such as multiple sclerosis. Inborn immunodeficiencies result from defects that interfere with innate, humoral, or cell-mediated defenses. **AIDS** is an acquired immunodeficiency caused by **HIV**.
- Antigenic variation, latency, and direct assault on the immune system allow some pathogens to thwart immune responses.

Is being infected with HIV the same as having AIDS? Explain.